Python Revision Assignment - 5th Semester AI Department

Instructions: Complete all questions in sequence. The difficulty level increases gradually. Use comments to explain your code wherever necessary.

1. Write a Python program to print 'Hello AI Department!' on the screen.

2. Create a Python program that takes two numbers as input and prints their sum, difference, product, and quotient.

3. Write a Python script to check if a given number is even or odd.

4. Create a program that uses a for loop to print all numbers from 1 to 50 that are divisible by 3,5 and 7.

5. Write a function that takes a string as input and returns the string reversed.

6. Write a Python program to count the number of vowels and consonants in a given string.

7. Create a Python list with 10 elements. Write code to find the largest and smallest number in the list.

8. Write a Python program to demonstrate the use of a dictionary. Create a dictionary of 5 students with their marks and print the student with the highest marks.

9. Write a program to read a text file and count how many times each word appears in it.

10. Write a Python class named 'Rectangle' with attributes length and width. Include methods to calculate area and perimeter.

11. Create a Python program that reads a list of numbers from the user and removes all duplicates while maintaining the order.

12. Write a Python function that takes a number as input and returns True if it is a prime number, otherwise False.

13. Using NumPy, create a 3x3 matrix and find its transpose and determinant.

14. Write a Python script that uses pandas to read a CSV file and display the first 5 rows of data.

15. Build a simple Python program that uses scikit-learn to train a linear regression model on a small dataset and predicts output for new input.